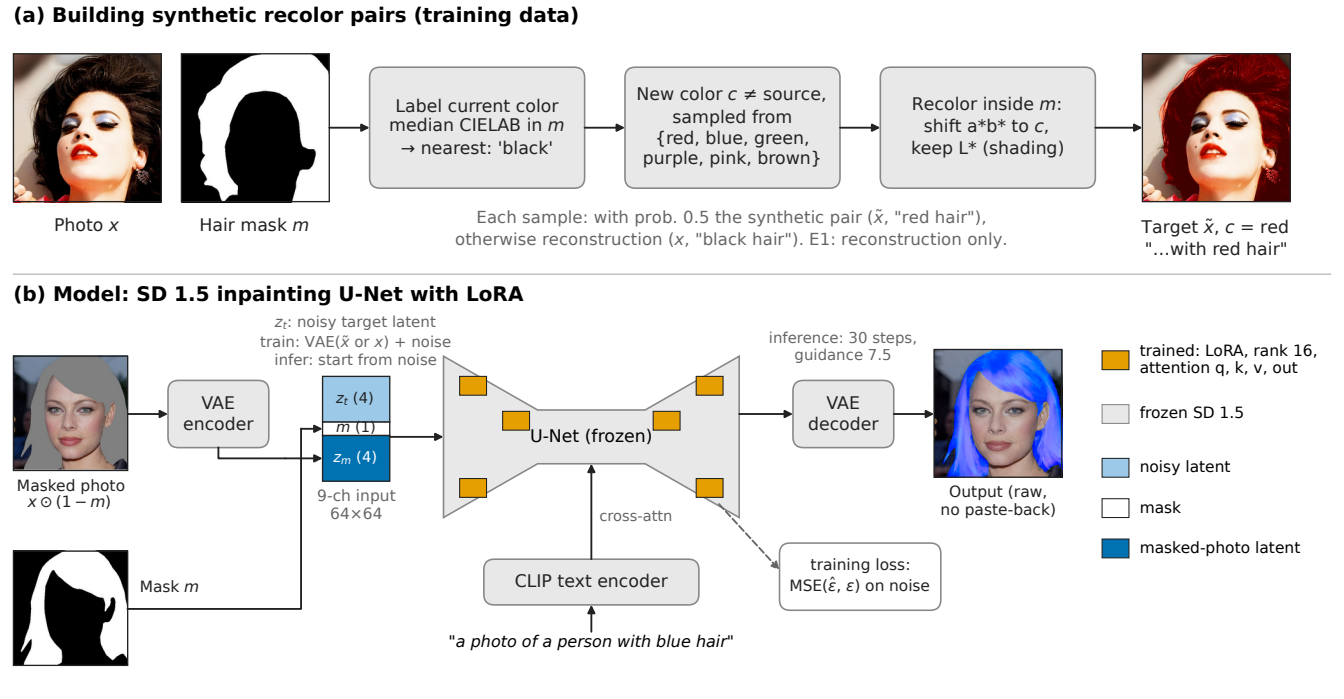

(a) Building synthetic recolor pairs (training data)
Photo $x$
Hair mask $m$
Label current color median CIELAB in $m$ → nearest: 'black'
New color $c \neq$ source, sampled from {red, blue, green, purple, pink, brown}
Recolor inside $m$: shift a*b* to $c$, keep L* (shading)
Target $\tilde{x}$, $c$ = red "...with red hair"
Each sample: with prob. 0.5 the synthetic pair ($\tilde{x}$, "red hair"), otherwise reconstruction ($x$, "black hair"). E1: reconstruction only.
(b) Model: SD 1.5 inpainting U-Net with LoRA
$z_t$: noisy target latent
train: VAE($\tilde{x}$ or $x$) + noise
infer: start from noise
Masked photo $x \odot (1-m)$
VAE encoder
$z_t$ (4)
$m$ (1)
$z_m$ (4)
9-ch input 64×64
U-Net (frozen)
cross-attn
CLIP text encoder
"a photo of a person with blue hair"
Mask $m$
inference: 30 steps, guidance 7.5
VAE decoder
Output (raw, no paste-back)
training loss: MSE($\hat{\varepsilon}$, $\varepsilon$) on noise
trained: LoRA, rank 16, attention q, k, v, out
frozen SD 1.5
noisy latent
mask
masked-photo latent